Study notes: Module 4, transient circuits and alternating currents

Murugeshan, *Electricity and Magnetism* (10th ed.), Ch. 12 §12.1 and §12.2 (Module 4, syllabus item 15–16). Figures are drawn for these notes from the book formulas; the book figure number is in each caption. Problems are not included.

Section 1: growth and decay of current in an L-R circuit (Murugeshan 12.1, 12.2)

12.1 Growth of current in a circuit with resistance and inductance

Source: book-sources/m4-s12-1-growth.html, Murugeshan pp. 312–313 (printed). Syllabus item 15. Equations (1)–(4) of §12.1 are given in full.

Before you start

- Ohm's law and the emf of a cell (Plus Two; pre-physics P1).
- Separating variables and integrating a first-order equation (Beyond; pre-math M4). Learn the steps here before the derivation.
- Exponentials and logarithms (pre-math M3).

1. The circuit

Take an inductance L and a resistance R in series with a cell of steady emf E. A key K closes the circuit (Fig. 12.1). When the key is pressed, the current does not jump to its final value. It grows from zero to a maximum I_0. Let I be the current at time t.

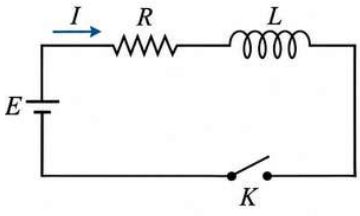


Fig. 12.1 (book): inductance L and resistance R in series with a cell E and key K. Schematic.

2. The loop equation

The inductor opposes the change of current with the back emf $\varepsilon = -L\,dI/dt$. Going round the loop, the cell emf is balanced by the drop across R and by the back emf of L:

$$E = RI + L\frac{dI}{dt} \qquad (12.1(1))$$

3. The steady current

When the current reaches its maximum I_0, it no longer changes, so $dI/dt = 0$ and the back emf is zero. Eq. (1) gives

$$E = RI_0 \qquad (12.1(2))$$

> **RECALL (PLUS TWO)** Ohm's law. With no change of current, the inductor is a plain wire, so the final current is $I_0 = E/R$. This is the same result the Plus Two circuit gives.

4. Separating the variables

Substitute $E = RI_0$ into Eq. (1). Then $RI_0 = RI + L\,dI/dt$, so

$$R(I_0 - I) = L\frac{dI}{dt} \quad \Rightarrow \quad \frac{dI}{I_0 - I} = \frac{R}{L}\,dt$$

The left side contains only I and the right side only t. The two can be integrated separately.

5. Integrating and fixing the constant

Integrating both sides,

$$-\log(I_0 - I) = \frac{R}{L}t + C \qquad (12.1(3))$$

where C is the constant of integration. At $t = 0$, $I = 0$, so $-\log I_0 = C$. Put this back into Eq. (3):

$$\log_e \frac{I_0 - I}{I_0} = -\frac{R}{L}t \quad \Rightarrow \quad 1 - \frac{I}{I_0} = e^{-(R/L)t}$$

6. The result

$$I = I_0\left(1 - e^{-(R/L)t}\right) \qquad (12.1(4))$$

Eq. (4) is the current in the L-R circuit at any instant. It starts at zero and tends to I_0 as t becomes large.

7. The time constant

The quantity L/R is the *time constant* τ of the circuit. Put $t = \tau = L/R$ in Eq. (4):

$$I = I_0\left(1 - e^{-1}\right) = 0.632\, I_0$$

So the time constant is the time the current takes to grow to 0.632 of its final value. A larger L/R means a slower growth. At $t = 2\tau$ the current is $0.865\, I_0$; at 3τ it is $0.950\, I_0$. Fig. 12.2 shows this for three values of L/R.

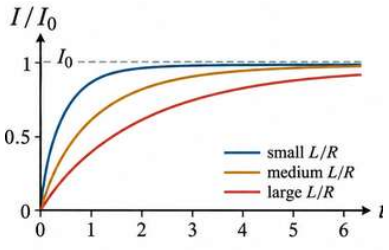


Fig. 12.2 (book): current I/I0 against time for low, medium and large L/R. Drawn from I = I0(1 − e^(−Rt/L)).

8. Worked example (printed p. 314)

An emf of 10 V is applied to a circuit with $R = 10\ \Omega$ and $L = 0.5$ H. The time to reach 63.2% of the final value is found from Eq. (4). Here $R/L = 20\ \text{s}^{-1}$, so $e^{-20t} = 0.368$ and $t = \frac{\ln 2.717}{20} = 0.05$ s. The time constant is $L/R = 0.05$ s, which is the same value. This is no accident: 63.2% is reached at $t = \tau$ by definition.

12.2 Decay of current in a circuit with inductance and resistance

Source: book-sources/m4-s12-2-decay.html, Murugeshan pp. 313–314 (printed), with Examples 1 and 2. Syllabus item 16. Equations (1) and (2) are given in full. Example 1 is a growth example; it is worked in §12.1 of these notes.

Before you start

- §12.1 of these notes: the loop equation and the time constant L/R.
- Exponentials and logarithms (pre-math M3).

1. What changes when the cell is removed

Once the current has reached I_0, the circuit is broken from the cell and closed on itself. The emf of the cell is now zero, so $E = 0$. The inductor still drives a current, because its back emf $-L\,dI/dt$ opposes the fall of current. The back emf keeps the current from dropping at once.

2. Loop equation with E = 0

$$0 = RI + L\frac{dI}{dt} \qquad (12.2(1))$$

Rearranging, $dI/I = -(R/L)\,dt$. The variables are separated again.

3. Integration

Integrate, with $\log_e I = -(R/L)t + C$. At $t = 0$, $I = I_0$, so $C = \log_e I_0$. Then

$$\log_e \frac{I}{I_0} = -\frac{R}{L}t \quad \Rightarrow \quad I = I_0 e^{-(R/L)t} \tag{12.2(2)}$$

4. Reading Eq. (2)

The current falls exponentially. The time constant $\tau = L/R$ is now the time for the current to fall to $1/e \approx 0.368$ of its initial value. The rate of decay is

$$\frac{dI}{dt} = -\frac{R}{L} I_0 e^{-(R/L)t} = -\frac{R}{L} I$$

The larger the ratio R/L, the more rapidly the current dies away (Fig. 12.4). The decay and growth curves are complementary (Fig. 12.5). The decay curve is the growth curve turned upside down.

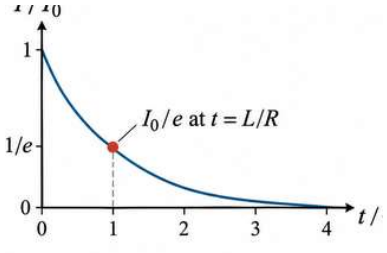


Fig. 12.3 (book): decay of current I = I0 e^(−Rt/L) against time. At t = L/R the current is I0/e.

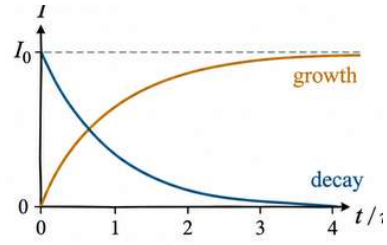


Fig. 12.5 (book): growth and decay curves shown together; they are complementary.

5. Worked example (printed p. 315)

An inductance of 500 mH and a resistance of 5 Ω are connected to a 10 V emf. The final current is $I_0 = E/R = 2$ A. When the cell is removed and the terminals are joined, $R/L = 10\ \text{s}^{-1}$ and $I = 2e^{-10t}$. After 0.05 s, $I = 1.213$ A. After 0.2 s, $I = 0.271$ A.

Section 2: charge in C-R and L-C circuits; leakage (Murugeshan 12.3 to 12.5)

12.3 Growth and decay of charge in a circuit with resistance and capacitance

Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 315–317 (printed), with Example 1 (capacitor charged through 2 MΩ) and Examples 2 and 3. Syllabus items 15 and 16. Equations (a)(1)–(a)(3) and (b)(1)–(b)(3) are given in full.

Before you start

- Capacitance and charge, $Q = CV$ (Plus Two; pre-physics P3).
- Ohm's law (pre-physics P1) and the first-order equation solved in §12.1 of these notes.

1. Growth of charge (charging)

> **RECALL (PLUS TWO)** The charge on a capacitor is $Q = CV$, so the p.d. across it is $V = Q/C$. In the loop below, the capacitor and the resistor share the emf, as in Plus Two series circuits.

A capacitor C and a resistance R are joined to a cell of emf E through a Morse key K (Fig. 12.6). When the key is pressed, a momentary current I flows through R. Let Q be the charge on the capacitor at time t. The p.d. across the capacitor is Q/C and the p.d. across the resistor is RI.

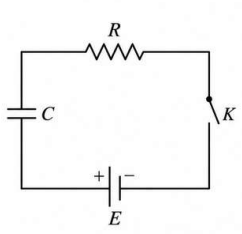


Fig. 12.6 (book): capacitor C, resistance R, Morse key K and cell E in series.

The emf equation of the circuit is

$$E = \frac{Q}{C} + RI \qquad (12.3(a)(1))$$

Since $I = dQ/dt$, this is $E = Q/C + R\,dQ/dt$.

2. The final charge

The charge grows until it reaches its maximum Q_0. At that instant $I = dQ/dt = 0$ and the p.d. across the capacitor equals the emf: $E = Q_0/C$.

3. Separating the variables

Substitute $E = Q_0/C$ into the equation, so $Q_0/C = Q/C + R\,dQ/dt$. Multiply by C:

$$(Q_0 - Q) = CR\frac{dQ}{dt}$$
$$\frac{dQ}{Q_0 - Q} = \frac{dt}{CR} \qquad (12.3(a)(2))$$

4. Integration and boundary condition

Integrating gives $-\log_e(Q_0 - Q) = t/CR + K$. At $t = 0$, $Q = 0$, so $K = -\log_e Q_0$. Then $\log_e \frac{Q_0 - Q}{Q_0} = -\frac{t}{CR}$, which gives

$$Q = Q_0\left(1 - e^{-t/CR}\right) \qquad (12.3(a)(3))$$

5. Time constant

The time constant is $\tau = CR$. At $t = CR$, $Q = Q_0(1 - e^{-1}) = 0.632\,Q_0$. The time constant is therefore the time for the charge to reach 0.632 of its maximum. The rate of growth is $dQ/dt = (Q_0/CR)e^{-t/CR}$. A smaller product CR gives faster charging. The rate of charging is greatest at the start and falls as Q approaches Q_0 (Fig. 12.7).

6. Decay of charge (discharging)

Now release the Morse key so that the capacitor, charged to Q_0, discharges through R. The charge flows out and forms a current. Here $E = 0$, so

$$R\frac{dQ}{dt} + \frac{Q}{C} = 0 \qquad (12.3(b)(1))$$

Separating the variables and integrating with $Q = Q_0$ at $t = 0$ gives

$$Q = Q_0 e^{-t/CR} \qquad (12.3(b)(2))$$

The charge decays exponentially and becomes zero only after an infinite time. The rate of discharge is

$$I = \frac{dQ}{dt} = -\frac{Q_0}{CR}e^{-t/CR} = -\frac{Q}{CR} \qquad (12.3(b)(3))$$

In Eq. (b)(2), at $t = CR$, $Q = Q_0 e^{-1} = 0.368\,Q_0$. So the time constant is also the time for the charge to fall to $1/e$ of its maximum. Growth and decay are shown together in Fig. 12.7.

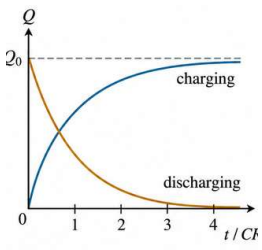


Fig. 12.7 (book): charge against time; growth to Q_0 and discharge from Q_0 are complementary.

7. Worked example 1 (printed p. 316)

A capacitor is charged through a resistance of $2\ \text{M}\Omega$. It takes 0.5 s to reach three quarters of its final charge. From Eq. (a)(3), $\frac{3}{4} = 1 - e^{-0.5/CR}$, so $e^{0.5/CR} = 4$ and $C = 0.5/(2 \times 10^6 \ln 4) = 0.18\ \mu\text{F}$.

8. Worked example 2 (printed p. 317)

A capacitor of $0.1\ \mu\text{F}$ is discharged through $10\ \text{M}\Omega$. The time for the p.d. to fall to half its value is $t = CR \ln 2 = 0.6931$ s, since $Q/Q_0 = V/V_0$ for the same capacitor.

12.4 Measurement of high resistance by leakage (background)

Background. *This section is outside the Module 4 items (15–22). It is kept because it uses the discharge law of §12.3 and the book gives it in full. No equations are numbered in the book for this section. Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 318–319 (printed).*

1. Principle

A capacitor of capacitance C and initial charge Q_0 is allowed to discharge through a high resistance R for a time t. From §12.3, the charge that remains is $Q = Q_0 e^{-t/CR}$. Therefore

$$\frac{Q_0}{Q} = e^{t/CR}$$

$$\Rightarrow$$

$$R = \frac{t}{C \log_e (Q_0/Q)} = \frac{t}{2.3026\, C \log_{10} (Q_0/Q)}$$

If R is high, CR is large and the discharge is very slow. Measuring Q_0/Q then gives R.

2. Circuit and procedure

C is a capacitor of known value, R is the high resistance, B.G. is a ballistic galvanometer, E is a cell and K_1, K_2, K_3 are tap keys (Fig. 12.8).

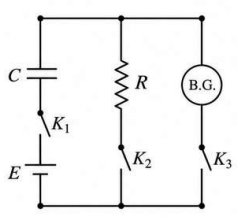


Fig. 12.8 (book): capacitor C, high resistance R, ballistic galvanometer B.G., cell E and keys K1, K2, K3.

(*i*) With K_2 and K_3 open, charge C fully through K_1. Open K_1 and close K_3. The capacitor discharges through the galvanometer, which gives a throw θ_0, proportional to Q_0.

(*ii*) Charge C again, open K_1, and close K_2 for a known time t. Some charge leaks through R. Open K_2 and close K_3. The remaining charge Q gives a throw θ, proportional to Q.

Since $Q_0/Q = \theta_0/\theta$,

$$R = \frac{t}{2.3026\, C \log_{10} (\theta_0/\theta)}$$

3. Graph

A plot of $\log_{10}(\theta_0/\theta)$ against t is a straight line through the origin (Fig. 12.9). Its slope gives $\frac{t}{\log_{10}(\theta_0/\theta)}$, and with C known, R follows.

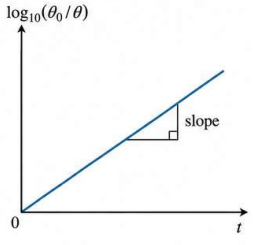


Fig. 12.9 (book): straight line of log10(θ0/θ) against t; the slope gives t / log10(θ0/θ).

4. Worked examples (printed pp. 318–319)

(1) For $C = 2\ \mu$F, $R = 100\ \text{M}\Omega$ and $Q_0/Q = 2$, the leakage time is $t = 2.3026\, CR \log_{10} 2 = 138.7$ s. (2) For $C = 0.2\ \mu$F, a kick of 12 cm and then 6 cm after 2 s gives $R = 14.43\ \text{M}\Omega$.

12.5 Charging and discharging of a capacitor through an inductance

Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 319–321 (printed). Syllabus items 15 and 16. The book numbers the charging equations (1)–(4) and, after a new heading, the discharge equations (1)–(4) again; these are tagged here as (i) and (ii). Section 12.6 onwards is outside the module and is not included.

Before you start

- The charge and the p.d. of a capacitor (pre-physics P3); self-inductance and back emf (pre-physics P2).
- Simple harmonic motion: $d^2x/dt^2 + \omega^2 x = 0$ (pre-math H2).

1. Charging a capacitor through an inductance

A capacitance C and an inductance L are joined to a cell E through a Morse key (Fig. 12.10). Pressing the key starts a momentary current through L and the capacitor begins to charge. Let Q be the charge on C at time t. The emf equation is

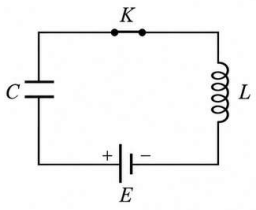


Fig. 12.10 (book): capacitor C and inductance L joined through a Morse key to a cell E.

$$L\frac{dI}{dt} + \frac{Q}{C} = E \qquad (12.5(i)(1))$$

Using $I = dQ/dt$, so that $dI/dt = d^2Q/dt^2$, and writing $Q_0 = CE$ for the maximum charge, the equation becomes

$$L\frac{d^2Q}{dt^2} + \frac{Q}{C} = E$$
$$\frac{d^2Q}{dt^2} + \frac{1}{LC}(Q - Q_0) = 0 \qquad (12.5(i)(2))$$

This is the equation of undamped simple harmonic motion, with period

$$T = \frac{2\pi}{\sqrt{1/LC}} = 2\pi\sqrt{LC} \qquad (12.5(i)(3))$$

The charging is oscillatory. The frequency is

$$\nu = \frac{1}{2\pi\sqrt{LC}} \qquad (12.5(i)(4))$$

2. Discharging a capacitor through an inductance

The capacitor is charged to Q_0 with the inductance out of the circuit. The key is then released so that C discharges through L (Fig. 12.11). No external emf acts during the discharge.

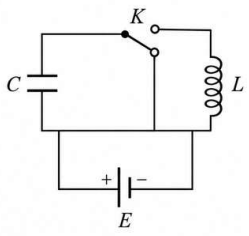


Fig. 12.11 (book): capacitor C discharging through inductance L with no external emf.

$$\frac{Q}{C} + L\frac{dI}{dt} = 0$$
$$\frac{d^2Q}{dt^2} + \frac{Q}{LC} = 0 \qquad (12.5(ii)(1))$$

This is undamped simple harmonic motion. Put $1/LC = \omega^2$. The general solution is

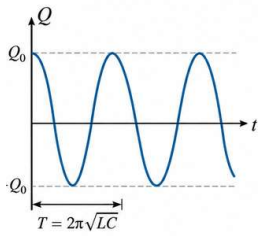
$$Q = A\sin(\omega t + \theta) \qquad (12.5(ii)(2))$$

with $\omega = 1/\sqrt{LC}$. At $t = 0$, $Q = Q_0$, which gives $Q_0 = A\sin\theta$. Taking $\theta = \pi/2$ gives $A = Q_0$, and

$$Q = Q_0 \sin\left(\frac{t}{\sqrt{LC}} + \frac{\pi}{2}\right) \qquad (12.5(ii)(3))$$

So Q decreases sinusoidally: the discharge is oscillatory. The natural frequency is

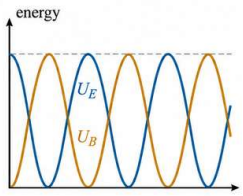
$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \qquad (12.5(ii)(4))$$



Undamped oscillation of charge, $Q = Q_0 \sin(t/\sqrt{LC} + \pi/2)$, period $T = 2\pi\sqrt{LC}$ (book, §12.5).

3. Energy in the oscillation

When C is fully charged, all the energy is stored in the electric field of the capacitor. During discharge, a current flows through the coil for a short time and the energy is stored in the magnetic field of the coil. The energy then passes back and forth between the two fields, which is the basis of the oscillations used in wireless telegraphy.



Energy alternates between the electric field of C and the magnetic field of L (book, §12.5).

Section 3: emf of a rotating coil; mean, r.m.s. and virtual values (Murugeshan 13.1)

13.1 EMF induced in a coil rotating in a magnetic field

Source: book-sources/m4-s3-ac-emf.html, Murugeshan pp. 329–333 (printed), with the impedance definitions at printed p. 333. Syllabus items 17 (and the peak, mean and r.m.s. values needed for items 18–22). The book does not number the equations of this section, so they are shown unnumbered.

Before you start

- Magnetic flux $\phi = BA\cos\theta$ and Faraday's law (pre-physics P4).
- Sinusoidal functions and their integrals (pre-physics P5; pre-math M5, M7).

1. The rotating coil

Take a rectangular coil of N turns, of length a and width b, rotating with uniform angular velocity ω about its axis in a uniform magnetic field $\mathbf{B}$ (Fig. 13.1). The axis is at right angles to the field. As the coil turns, the flux through it changes, so an emf is induced.

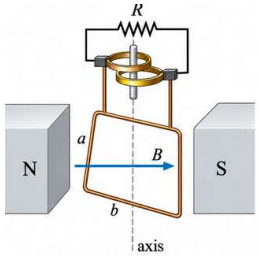


Fig. 13.1 (book): rectangular coil rotating between poles N and S, with slip rings and resistance R.

Start timing when the plane of the coil is at right angles to $\mathbf{B}$, that is when the normal to the coil makes zero angle with $\mathbf{B}$. At time t the normal makes angle $\theta = \omega t$ with $\mathbf{B}$. The flux linked with N turns is

$$\phi = NBA\cos\theta = NBA\cos\omega t, \qquad A = ab$$

2. The induced emf

> **RECALL (PLUS TWO)** The flux through a coil is $\phi = BA\cos\theta$, and the induced emf is $-d\phi/dt$ (Faraday). Here the coil has N turns and $\theta = \omega t$, which gives the sinusoid below.

By Faraday's law, the instantaneous emf is

$$E = -\frac{d\phi}{dt} = NBA\omega \sin \omega t = E_0 \sin \omega t$$

where $E_0 = NBA\omega$ is the *peak value* of the emf. With $\omega = 2\pi\nu$, the emf is a sine curve (Fig. 13.2). A coil in a closed circuit carries an *alternating current* $I = I_0 \sin \omega t$.

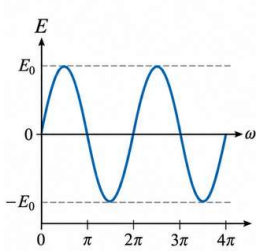


Fig. 13.2 (book): emf E = E_0 sin ωt against ωt; a sine curve.

3. One cycle

The time for one cycle is the *time period* $T = 1/\nu$. The peak value of the current or voltage is the *amplitude* (Fig. 13.3).

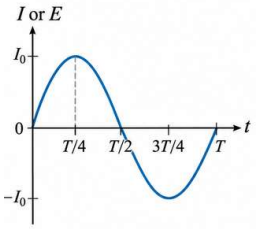


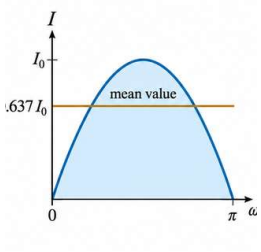
Fig. 13.3 (book): one cycle of ac voltage or current, with peak value and the times T/4, T/2, 3T/4, T.

4. Mean value

The mean value is the average over half a cycle:

$$\begin{aligned} I_{mean} &= \frac{\int_0^{\pi/\omega} I_0 \sin \omega t \, dt}{\pi/\omega} \\ &= \frac{2I_0}{\pi} \\ &= 0.637\, I_0 \end{aligned}$$

Similarly $E_{mean} = 0.637\, E_0$. The mean is the area under the half-cycle curve divided by its base (Fig. 13.3 shading).



Mean value over half a cycle, 0.637 I_0 (book, §13.1, shaded area).

5. Root mean square (r.m.s.) value

The r.m.s. value is the square root of the average of I^2 over a complete cycle:

$$\overline{I^2} = \frac{\int_0^{2\pi/\omega} I_0^2 \sin^2 \omega t \, dt}{2\pi/\omega} = \frac{I_0^2}{2}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707\, I_0$$

Similarly $E_{rms} = E_0/\sqrt{2} = 0.707\, E_0$.

6. Virtual (effective) value

The r.m.s. value is also called the *virtual* or *effective* value. It is the direct current that would produce the same heat in the same resistance in the same time. Equating the heat from $I = I_0 \sin \omega t$ to $I_v^2 RT$ over one period gives $I_v = I_0/\sqrt{2}$ again.

$$I_v = \frac{I_0}{\sqrt{2}} = I_{rms}$$

7. Form factor

The form factor measures the shape of the wave. It is the ratio of the r.m.s. value to the mean value:

$$\begin{aligned} \text{Form factor} &= \frac{I_{rms}}{I_{mean}} \\ &= \frac{0.707\, E_0}{0.637\, E_0} \\ &= 1.11 \end{aligned}$$

8. Mean emf over a half cycle

With $E = E_0 \sin \omega t$, integrating over half a period $T/2$ gives $E_{av} = 2E_0/\pi$. The r.m.s. emf is obtained from the mean of E^2 over one period, using $\int_0^T \sin^2 \omega t \, dt = T/2$, which gives $E_{rms} = E_0/\sqrt{2}$.

9. Impedance, reactance and admittance

In an AC circuit the current is controlled by resistance, inductance and capacitance. The *impedance* Z is the ratio of the applied emf to the current, $Z = E/I$. It is the hindrance the circuit offers to the alternating

current. The part of the hindrance due to inductance and capacitance is the *reactance*. Inductive reactance is X_L and capacitive reactance is X_C. The inverse of the impedance is the *admittance* $Y = 1/Z$.

Section 4: AC circuits with R, L and C; phasors and the j operator (Murugeshan 13.2)

13.2 Alternating current circuit theory

Source: book-sources/m4-s4-ac-circuits.html, Murugeshan pp. 334–337 (printed). Syllabus items 18, 19 and 20 (R, L and C alone; the L-R and C-R circuits; the j operator). The parallel resonant circuit, §13.2(vi), is in Section 5 with §13.4. Equations are tagged with the book's numbers, (i) to (v).

Before you start

- Phase of a sinusoid; lag and lead (pre-physics H2).
- Complex numbers, $j^2 = -1$, and the phasor (Beyond; pre-math B1, B2). Each step with j is explained below.

1. (i) A pure resistance

> **RECALL (PLUS TWO)** Ohm's law $V = IR$ (Plus Two). For a pure resistance, E/R gives the current directly, and the current is in phase with the emf. The same law holds for the complex form $E = E_0 e^{j\omega t}$.

Let $E = E_0 e^{j\omega t}$ be applied across a pure resistance R (Fig. 13.4). The current is

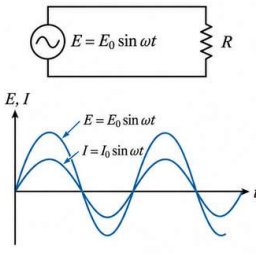


Fig. 13.4 and 13.5 (book): pure resistance; E and I in phase.

$$I = \frac{E}{R} = \frac{E_0 e^{j\omega t}}{R} = I_0 e^{j\omega t}$$
$$I_0 = \frac{E_0}{R}$$

The actual current is the imaginary part of $I_0 e^{j\omega t}$, which is $I_0 \sin \omega t$. The current is *in phase* with the voltage (Fig. 13.5).

2. (ii) A pure inductance

Let $E = E_0 e^{j\omega t}$ across an inductance L (Fig. 13.6). The voltage across L is $L\,dI/dt$. Integrating $L\,dI/dt = E_0 e^{j\omega t}$ gives

$$I = \frac{E}{j\omega L} = \frac{E_0}{\omega L} e^{j(\omega t - \pi/2)} \qquad (13.2(\text{ii})(1))$$

With $I_0 = E_0/\omega L$, the actual current is

$$I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right) \qquad (13.2(\text{ii})(2))$$

In a pure inductor the current *lags* the emf by $\pi/2$ (Fig. 13.7). Here ωL is the inductive reactance, and $j\omega L$ is the impedance of L: the voltage leads the current by $90°$.

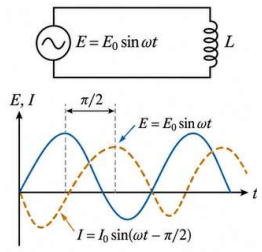


Fig. 13.6 and 13.7 (book): pure inductance; current lags the emf by π/2.

3. (iii) A pure capacitance

Let $E = E_0 e^{j\omega t}$ across a capacitor C (Fig. 13.8). With charge q, $q/C = E$. The current is $I = dq/dt = j\omega C E$, so

$$I = \frac{E_0}{1/\omega C} e^{j(\omega t + \pi/2)} = I_0 e^{j(\omega t + \pi/2)}$$
$$I_0 = \frac{E_0}{1/\omega C}$$

The actual current is $I_0 \sin(\omega t + \pi/2)$. In a pure capacitor the current *leads* the emf by $\pi/2$ (Fig. 13.9). The capacitive reactance is $1/\omega C$, and its impedance is $1/(j\omega C) = -j/(\omega C)$.

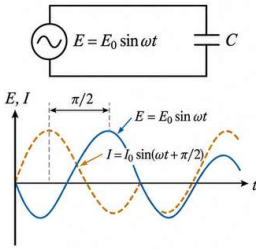


Fig. 13.8 and 13.9 (book): pure capacitance; current leads the emf by π/2.

4. (iv) Inductance and resistance in series

Let $E = E_0 e^{j\omega t}$ be applied to L and a non-inductive R in series (Fig. 13.10). The impedance is $Z = R + j\omega L$. The potential drops are $j\omega LI$ across L and RI across R, so $E = j\omega LI + RI$ and

$$I = \frac{E}{R + j\omega L} \qquad (13.2(\text{iv})(1))$$

$$I = \frac{E}{Z} \qquad (13.2(\text{iv})(2))$$

$$Z = R + j\omega L \qquad (13.2(\text{iv})(3))$$

Write Z in polar form, $Z = \sqrt{R^2 + \omega^2 L^2}\, e^{j\theta}$ with $\tan\theta = \omega L/R$. Then

$$I = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}}\, e^{j(\omega t - \theta)} \qquad (13.2(\text{iv})(4))$$

$$I = I_0 e^{j(\omega t - \theta)} \qquad (13.2(\text{iv})(5))$$

$$I_0 = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}} \qquad (13.2(\text{iv})(6))$$

I_0 is the peak current. The impedance is $Z = \sqrt{R^2 + \omega^2 L^2}$. The current *lags* the emf by $\theta = \tan^{-1}(\omega L/R)$ (Fig. 13.11).

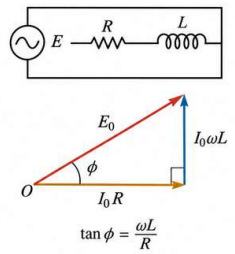


Fig. 13.10 (book): R and L in series, with the phasor diagram of I_0R and $I_0\omega L$.

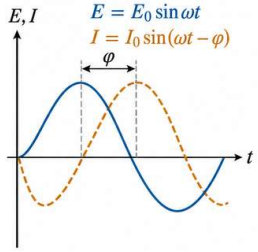


Fig. 13.11 (book): emf and current for R and L in series; current lags by θ.

5. (v) Capacitance and resistance in series

Let $E = E_0 e^{j\omega t}$ be applied to R and C in series (Fig. 13.12). Let I be the current and q the charge on C. Then

$$I\left(R + \frac{1}{j\omega C}\right) = E = E_0 e^{j\omega t} \qquad (13.2(\text{v})(1))$$

$$I = \frac{E}{R + 1/(j\omega C)} \qquad (13.2(\text{v})(2))$$

$$I = \frac{E}{Z} \qquad (13.2(\text{v})(3))$$

The impedance is $Z = R + 1/(j\omega C) = R - j/(\omega C)$. Writing Z in polar form and taking $\theta = \tan^{-1}[(1/\omega C)/R]$,

$$I = \frac{E}{R - j/(\omega C)} = \frac{E_0 e^{j\omega t}}{\sqrt{R^2 + 1/(\omega C)^2}}\, e^{-j\theta} \qquad (13.2(\text{v})(4))$$

$$\theta = \tan^{-1}\left(\frac{1/\omega C}{R}\right) \qquad (13.2(\text{v})(5))$$

$$I = \frac{E_0}{\sqrt{R^2 + (1/\omega C)^2}}\, e^{j(\omega t + \theta)} \qquad (13.2(\text{v})(6))$$

The current *leads* the applied voltage by θ (Fig. 13.13). The actual current is $I = I_0 \sin(\omega t + \theta)$.

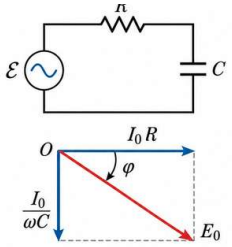


Fig. 13.12 (book): R and C in series, with the phasor diagram.

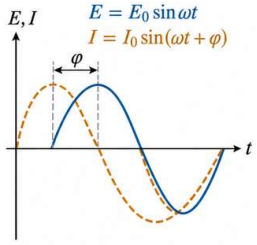


Fig. 13.13 (book): emf and current for R and C in series; current leads by θ.

6. The j operator as a rotation

$j = \sqrt{-1}$ turns a phasor through $90°$ anticlockwise. $-j$ turns it $90°$ clockwise. In a circuit, E_L lies $90°$ ahead of I and E_C lies $90°$ behind it. Hence the total emf of a circuit with R, L and C is $E = E_R + jE_L - jE_C$ (Fig. 13.20). A complex impedance $Z = R + jX$ has resistance R and effective reactance $X = X_L - X_C$.

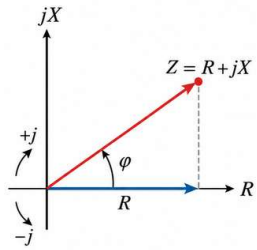


Fig. 13.20 (book): phasors E_R, jE_L and −jE_C; j rotates a phasor by 90°.

Section 5: series LCR and parallel resonance (Murugeshan 13.2(vi), 13.3, 13.4)

13.2 (vi), 13.3 and 13.4 Parallel and series resonance

Source: book-sources/m4-s5-series-parallel.html, Murugeshan pp. 338–346 (printed), with Examples 1 and 2 of §13.3 and Example 1 of §13.4. Syllabus items 20 (parallel L and C) and 21 (series LCR). The book numbers the series derivation by two methods; the equations are tagged (a) for the first and (b) for the second method.

Before you start

- The impedance and phase of R, L and C in series (Section 4 of these notes).
- Complex numbers, the conjugate and the modulus (pre-math B1, B3).

1. Parallel resonant circuit (§13.2(vi))

Take an inductance L and a capacitance C in parallel across an alternating emf $E = E_0 \sin \omega t$. The current in L lags the emf by $\pi/2$, so $I_L = (E_0/\omega L)\sin(\omega t - \pi/2)$. The current in C leads the emf by $\pi/2$, so $I_C = E_0/(1/\omega C)\sin(\omega t + \pi/2)$ (Fig. 13.14). The total current is

$$\begin{aligned} I &= I_L + I_C \\ &= E_0\left(\omega C - \frac{1}{\omega L}\right)\cos \omega t \end{aligned} \qquad (13.2(\text{vi})(1))$$

If at a frequency ν_0 we have $\omega C = 1/\omega L$, the current is zero. The circuit then offers an infinite impedance: it is a *parallel resonant circuit*. Its resonant frequency is

$$\nu_0 = \frac{1}{2\pi\sqrt{LC}} \qquad (13.2(\text{vi})(2))$$

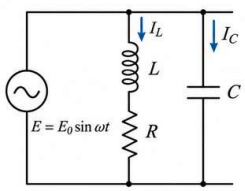


Fig. 13.14 (book): parallel L and C across E = E_0 sin ωt.

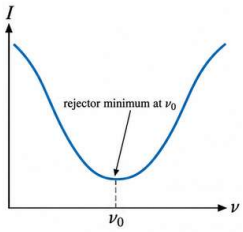


Fig. 13.15 and 13.23 (book): current against frequency, minimum at the parallel resonant frequency.

2. Series LCR circuit: first method (§13.3)

> **RECALL (PLUS TWO)** In Plus Two, resistances in series add, and the total opposition to current is the impedance. Here the reactances X_L and X_C subtract, so the impedance is $\sqrt{R^2 + (X_L - X_C)^2}$.

Let $E = E_0 \sin \omega t$ be applied to R, L and C in series (Fig. 13.16). Let I be the current and Q the charge on C. The emf equation is $L\,dI/dt + RI + Q/C = E_0 \sin \omega t$. Differentiating with respect to t gives

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{1}{C}\frac{dQ}{dt} = E_0\omega\cos\omega t \quad (13.3(a)(1))$$

Try the solution $I = I_0 \sin(\omega t - \phi)$, with I_0 and ϕ constants. Substituting I, dI/dt and d^2I/dt^2 and equating the coefficients of $\sin(\omega t - \phi)$ and $\cos(\omega t - \phi)$ gives

$$I = I_0 \sin(\omega t - \phi) \quad (13.3(a)(2))$$

$$\left(-L\omega^2 + \frac{1}{C}\right) I_0 = -E_0\omega\sin\phi \quad (13.3(a)(3))$$

$$R\omega I_0 = E_0\omega\cos\phi \quad (13.3(a)(4))$$

Dividing and squaring-and-adding give

$$\tan\phi = \frac{\omega L - 1/\omega C}{R} \quad (13.3(a)(5))$$

$$I_0 = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} \quad (13.3(a)(6))$$

$$I = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} \sin(\omega t - \phi) \quad (13.3(a)(7))$$

The quantity $\sqrt{R^2 + (\omega L - 1/\omega C)^2}$ is the impedance Z, which is $\sqrt{R^2 + (X_L - X_C)^2}$. The current lags the emf if $X_L > X_C$, leads it if $X_L < X_C$, and is in phase when $X_L = X_C$.

3. Series resonance

The impedance is least when $\omega L = 1/\omega C$, so $Z = R$ and the current is largest. This frequency is the resonant frequency $\nu_0 = 1/(2\pi\sqrt{LC})$. The largest current is $I_0 = E_0/R$. The sharpness of the peak depends on R: small R gives a sharp peak (Figs. 13.17, 13.18). The circuit is an *acceptor* circuit, because it admits the current of its own frequency most readily.

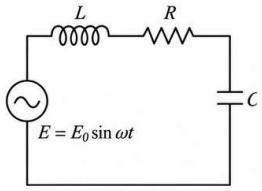
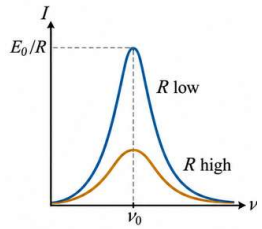


Fig. 13.16 (book): R, L and C in series across E = E_0 sin ωt.



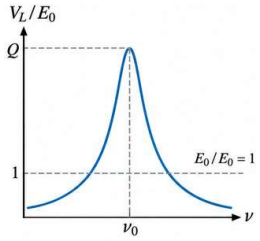
Figs. 13.17 and 13.18 (book): current against frequency at resonance, for low and high R.

$$I = \frac{E_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{C\omega}\right)^2}} \times \sin(\omega t - \phi) \quad (13.3(res)(1))$$

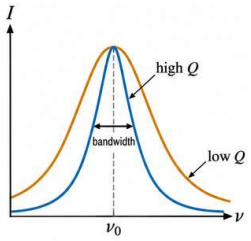
4. Voltage magnification and the Q-factor

At resonance $I_0 = E_0/R$, and the voltage across L is $V_L = X_L I_0 = (\omega_0 L/R)E_0 = QE_0$, where $Q = \omega_0 L/R$ is the quality factor. Thus the voltage across L (and across C) is Q times the applied voltage. This is the *voltage magnification*. It does not increase the power, because the reactive power is wattless. A larger Q gives a sharper, narrower

resonance curve (Fig. 13.18). The sharpness is measured by the bandwidth.



Voltage magnification at resonance: V_L / E₀ = Q (book, §13.3).



Sharp and flat resonance curves; a large Q gives a narrow bandwidth (book, §13.3).

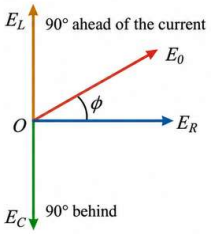


Fig. 13.19 (book): vector diagram of the series LCR circuit.

5. Vector diagram

In a series circuit the current is the same through R, L and C. The drop $E_R = I_0 R$ is in phase with the current, $E_L = \omega L I_0$ is $90°$ ahead of it, and $E_C = I_0/\omega C$ is $90°$ behind it (Fig. 13.19). The resultant of E_L and E_C is $E_L - E_C$, and $E_0^2 = E_R^2 + (E_L - E_C)^2$.

6. Series LCR circuit: second method (complex impedance)

Write $Z = Z_R + Z_L + Z_C = R + j(\omega L - 1/\omega C)$ (Fig. 13.21). With $E = E_0 e^{j\omega t}$,

$$E = E_0 e^{j\omega t} \qquad (13.3(b)(1))$$

$$Z = \sqrt{R^2 + (\omega L - 1/\omega C)^2}\, e^{j\phi}, \quad \tan\phi = \frac{\omega L - 1/\omega C}{R} \qquad (13.3(b)(2))$$

$$I = \frac{E}{Z} = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}\, e^{j(\omega t - \phi)} \qquad (13.3(b)(3))$$

$$I = I_0 e^{j(\omega t - \phi)} \qquad (13.3(b)(4))$$

$$i = \mathrm{Im}(I) = \frac{E_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} \sin(\omega t - \phi) \qquad (13.3(b)(5))$$

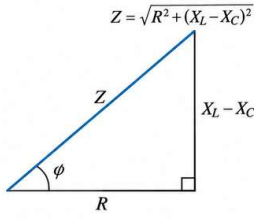
The actual current is the imaginary part of the complex current. The current lags the voltage by ϕ.

7. Parallel resonant circuit: the rejector (§13.4)

Let C be in parallel with the series combination of R and L, across $E = E_0 e^{j\omega t}$ (Fig. 13.22). The impedance of the L branch is $Z_1 = R + j\omega L$ and of the C branch is $Z_2 = 1/(j\omega C)$. Their parallel combination has

$$\frac{1}{Z} = \frac{R}{R^2 + (L\omega)^2} + j\left[C\omega - \frac{L\omega}{R^2 + (L\omega)^2}\right]$$

The admittance is least, and so the impedance greatest, at $\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$. If R is small, $\nu_0 = 1/(2\pi\sqrt{LC})$. At this frequency the impedance is $Z = L/(RC)$ and the current is least. A parallel circuit at resonance is a *rejector*: it blocks the current of its natural frequency (Fig. 13.23). In a parallel circuit, resonance gives maximum impedance; in a series circuit, minimum.



Impedance triangle for series LCR: Z = √(R² + (X_L − X_C)²) (book, §13.3).

8. Worked examples

§13.3, Example 1 (printed p. 344). $R = 50 + 2 = 52\ \Omega$, $L = 0.3$ H, $C = 40\ \mu$F, $\omega = 100\pi$, $E_{rms} = 200$ V. $X_L = 94.2\ \Omega$, $X_C = 79.6\ \Omega$, $Z = 53.97\ \Omega$, $I_{rms} = 3.71$ A. $\tan\phi = 0.28$, $\phi \approx 15°39'$, $\cos\phi = 0.964$, apparent power 742 VA, true power 716 W.

§13.3, Example 2 (printed p. 344). For resonance with $L = 5$ H and $\omega = 100\pi$, $C = 2\ \mu$F. The current at resonance is $I_{rms} = 100/100 = 1.0$ A. The p.d. across R is 100 V, across L 1570 V, and across C 1570 V, which are equal and opposite in phase.

§13.4, Example 1 (printed p. 347). $L = 2$ mH, $R = 15\ \Omega$, $C = 0.001\ \mu$F. The resonant frequency is $\nu_0 = 112\,590$ Hz (or $112\,596$ Hz by the approximation). The peak make-up current is $E_0 CR/L = 15 \times 10^{-6}$ A.

Section 6: power in AC circuits (Murugeshan 13.5)

13.5 Power in an AC circuit with resistance, inductance and capacitance

Source: book-sources/m4-s6-power.html, Murugeshan pp. 347–351 (printed), with Examples 1–5 of §13.5. Syllabus item 22. §13.6 (choke coil) begins on printed p. 351 and continues beyond this module; its heading is shown and its text is not included. Equations (1) and (2) are the book's; the Example 2 equations are tagged (ex2) with their book numbers.

Before you start

- Power in a DC circuit, $P = VI = I^2R$ (pre-physics P8).
- Products of sinusoids and the average of $\cos^2$ over a cycle (pre-math M5, M7).
- Phase angle ϕ between current and emf (Section 4 and 5 of these notes).

1. Instantaneous power

> **RECALL (PLUS TWO)** In a DC resistive circuit, $P = VI = I^2R = V^2/R$ (Plus Two). In an AC circuit the same formula holds for the instantaneous power EI, and the average is taken over a cycle with the factor $\cos\phi$.

Let the voltage and current be $E = E_0 \sin\omega t$ and $I = I_0 \sin(\omega t - \phi)$, where ϕ is the phase difference. Both vary with time, so the power EI is calculated at each instant and then averaged over a cycle. Using $\sin A \sin B = \tfrac{1}{2}[\cos(A-B) - \cos(A+B)]$,

$$\begin{aligned} E \times I &= E_0 I_0 \sin\omega t \, \sin(\omega t - \phi) \\ &= \frac{1}{2} E_0 I_0 \left[\cos\phi - \cos(2\omega t - \phi)\right] \end{aligned} \qquad (13.5(1))$$

2. Average power

Averaging over one complete period T, the $\cos(2\omega t - \phi)$ term integrates to zero, and

$$P = \frac{1}{T}\int_0^T EI\,dt = \frac{1}{2} E_0 I_0 \cos\phi \qquad (13.5(2))$$

Since $E_{rms} = E_0/\sqrt{2}$ and $I_{rms} = I_0/\sqrt{2}$,

$$P = E_{rms}\, I_{rms} \cos\phi$$

The product $E_{rms} I_{rms}$ is the *apparent power*. The factor $\cos\phi$ is the *power factor*. The true power is the apparent power times the power factor. The power factor is the ratio of the true power to the apparent power. For a circuit of R, L and C in series,

$$\tan\phi = \frac{\omega L - 1/\omega C}{R}$$

$$\cos\phi = \frac{R}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

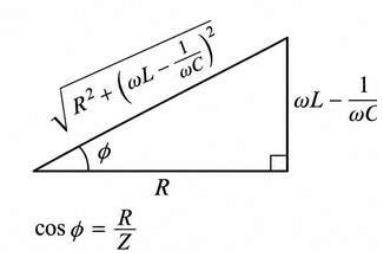


Fig. 13.24 (book): power-factor triangle with sides R, ωL − 1/ωC and Z.

3. Special cases

- Purely resistive: $\phi = 0$, $\cos\phi = 1$, and the true power is $E_v I_v$.
- Purely inductive: $\phi = 90°$, $\cos\phi = 0$; the true power is zero.
- Purely capacitive: $\phi = -90°$, $\cos\phi = 0$; the true power is zero.
- R and L in series: $\cos\phi = R/\sqrt{R^2 + (\omega L)^2}$.
- R and C in series: $\cos\phi = R/\sqrt{R^2 + 1/(\omega C)^2}$.

4. Wattless current

The current is *wattless* when the average power is zero. In a purely inductive or purely capacitive circuit with no resistance, $\phi = \pi/2$ and $\cos\phi = 0$. The current does no useful work; it is the *idle* current. This is the principle of the choke coil, which offers reactance to the current but absorbs no power. Splitting the current into components along and perpendicular to the emf (Fig. 13.25), the component $I\cos\phi$ gives the power, and $I\sin\phi$ is wattless.

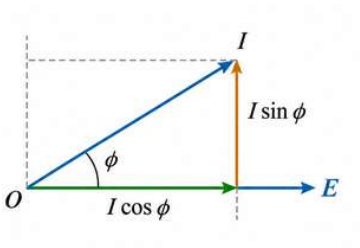
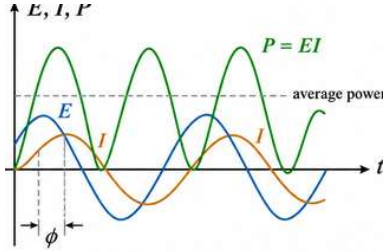


Fig. 13.25 (book): I cos φ (power component) and I sin φ (wattless component).



Emf, current and instantaneous power against time; the average power is shown dashed.

5. Worked examples

Example 1 (printed p. 348). A choke of inductance 5 H and resistance 200 Ω is connected to 10 V at 100 Hz. $\cos\phi = 200/\sqrt{200^2 + (2\pi\cdot 100\cdot 5)^2} = 0.062$. The power absorbed is 0.00189 W.

Example 2 (printed pp. 348–349). A 50 V r.m.s. source feeds an air-cored coil. At 50 Hz the power is 100 W and at 100 Hz it is 50 W. The power is dissipated only in R: $\frac{50^2}{R^2+(100\pi)^2L^2}R = 100$ and $\frac{50^2 R}{R^2+(200\pi)^2L^2} = 50$. Eliminating gives $R^2 = 2\times 10^4\pi^2L^2$, so $R = 100\sqrt{2}\,\pi L$, and then $R = 16.7\ \Omega$, $L = 37.3$ mH.

$$\frac{50^2}{[R^2+(100\pi)^2L^2]}R = 100\ \text{W} \qquad (13.5(\text{ex2})(1))$$

$$\frac{50^2 R}{[R^2+(200\pi)^2L^2]} = 50\ \text{W} \qquad (13.5(\text{ex2})(2))$$

$$2500\,R = 100\left[R^2 + 10^4\pi^2L^2\right] \qquad (13.5(\text{ex2})(3))$$

$$2500\,R = 50\left[R^2 + 4\times 10^4\pi^2L^2\right] \qquad (13.5(\text{ex2})(4))$$

$$R^2 = 2\times 10^4\times\pi^2L^2 \qquad (13.5(\text{ex2})(5))$$
$$R = 100\sqrt{2}\,\pi L$$

Example 3 (printed pp. 350–351). A coil of inductance 0.7 H in series with 50 Ω is connected to 200 V at 50 Hz. $Z = 225.6\ \Omega$, $I_{rms} = 0.89$ A. The power component is $I\cos\phi = 0.20$ A and the wattless component is $I\sin\phi = 0.87$ A.

Example 4 (printed p. 351). A coil of 0.1 H and 12 Ω on 220 V, 50 Hz: $X_L = 31.43\ \Omega$, $Z = 33.6\ \Omega$, and the wattmeter reading is $(220)^2/33.6\times 12/33.6 = 514.5$ W.

Example 5 (printed p. 351). A purely inductive circuit dissipates no power. The energy drawn from the source in one part of the cycle to build the magnetic field is returned to the source in the next part. The inductance stores energy and gives it back; only the resistance of the coil loses energy as heat.

6. Choke coil (§13.6, heading only)

The choke coil is an inductance used to control current without power loss. Its section begins on printed p. 351 and continues beyond this module. It is not included in these notes.